Six-sum kernel B/B2: direct quartic root solve (resolvent cubic)

Class: engineering; stage diagnostic of relation extraction. No end-to-end `S`, no rho comparison, no key-recovery or deployed-curve claim. **Date:** 2026-10-18. **Branch:** wt/t04. **Host:** shared 4-vCPU container, load about 2, no isolation tier earned; wall times are secondary, operation counts are exact.

What ran

`koblitz_sixsum_extract_probe` (report version 4), Koblitz K_0 $n = 41$ ($r \approx 2^{39}$), seed 20261012, $K \in \{10, 20\}$, 32 verified relations per cell, kernels 4sum and 6sum_b2_s4_strided, `KIC_S4_ROOTS=split` (control, unchanged) against `resolvent` (new), `--solver-bench=3000` (3000 complete quartics, best of 7 interleaved repeats). Every relation is re-added in the group: 0 rejects, 0 lift failures, 0 `s4_fallbacks`; both solvers agree on all benchmarked quartics. Raw records: `raw_resolvent_20261018/n41_k10_k20_{split,resolvent}.json` and `.stderr.txt`; frozen summary `resolvent_summary.json`.

Measured

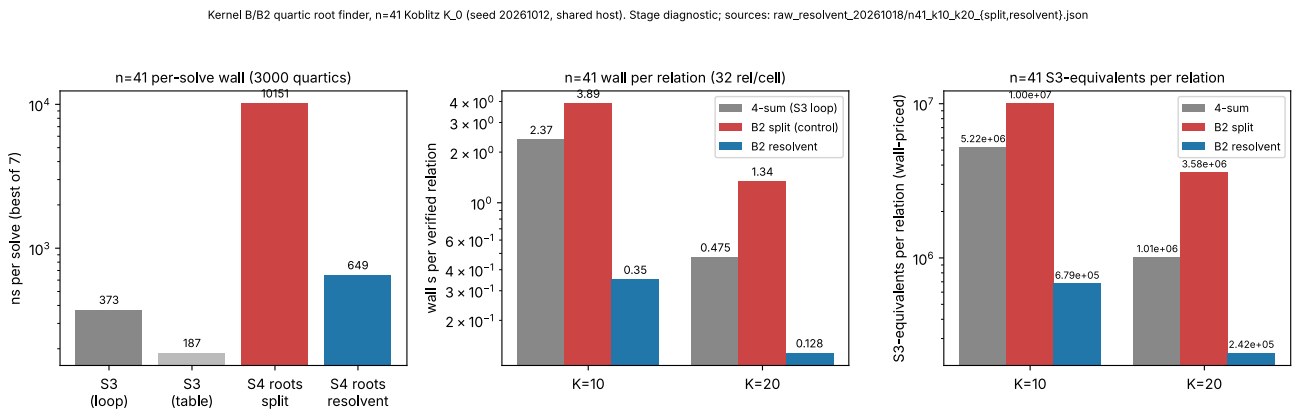


Figure 1: Left: per-solve wall at $n = 41$ (resolvent-run values; the split run measured S3 354 ns, roots split 9.54 us, resolvent 0.611 us). Middle: wall per verified relation. Right: S3-equivalents per relation, S4 solves priced by the measured S4-pair wall cost in S3 units. Sources: `raw_resolvent_20261018/n41_k10_k20_*.json`.

measured	4-sum	B2 split	B2 resolvent	unit
K=10 wall	2.37	3.89	0.350	s / relation
K=20 wall	0.475	1.34	0.128	s / relation
K=10 S3-equiv.	5.22e6	1.00e7	6.79e5	S3 / relation
K=20 S3-equiv.	1.01e6	3.58e6	2.42e5	S3 / relation
roots per solve	-	9.54 us	0.611 us	n=41, split run
field ops per quartic	-	1710 mul, 395 sqr, 7 inv	70 mul, 49 sqr, 2 inv, 8 table maps	counted

S4 solves per relation are identical across split and resolvent (3.71e5 at K=10, 1.32e5 at K=20); only the root finder changes.

Derived (from the table): resolvent against split, 11.1x (K=10) and 10.5x (K=20) less wall per relation; against the 4-sum, 6.8x and 3.7x. The resolvent root solve alone costs 1.73 S3 by wall and 2.19 by priced field operations (split run); an S4 pair with its coefficients costs 1.83 S3 by wall and 2.51 priced (resolvent run), against the break-even of about 12 S3 named on 2026-10-12.

Estimate, not measured end to end: the 4-sum above uses the crate's loop half-trace S3 (about 373 ns). A 4-sum using the byte-table half-trace S3 (about 187 ns in the same run) would be roughly 2x cheaper, which would reduce the B2 resolvent margin to about 3.4x (K=10) and 1.9x (K=20).

Scope and limits

Before (2026-10-12): one S4 root solve cost 20-60 S3 and B2 was 1-8x more S3-work than the 4-sum. After, on this ladder only: B2 with the resolvent solver is below the 4-sum at both K. No factor-base, linear-algebra or descent cost is included; $n = 53$ and an end-to-end cold `S` against the frozen baseline are not measured. $n = 23$ and 31 cells are saturated (a few S3 solves per relation) and give solver costs only (`raw_resolvent_20261018/n23_31_*`). Distance to matched rho: not measured here.

Reproduce

`KIC_S4_ROOTS=resolvent koblitz_sixsum_extract_probe --n=41 --a=0 --columns=10,20 --kernels=4sum,b2 --relations=32 --seed=20261012 --max-targets=4096 --max-s3=400000000 --max-s4=50000000 --cell-wall-s=1200 --solver-bench=3000`; graph: `plot_resolvent.py`.